

Data-Driven Framework for Assessing Water Potability and Enhancing Public Health Security

Euphoria

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**When you turn on the faucet at home to
get a glass of water, do you actually
know if it's safe, or are you just hoping
that it is?**



SDG Alignment

Primary SDG:



SDG 6: Clean Water and Sanitation

Specific Problem:



Despite national efforts, a significant portion of the Philippine population remains vulnerable to poor water quality. Companies and local water districts often provide data on water supply, but the actual "potability" at the point of consumption is frequently questioned due to aging infrastructure and environmental contamination. There is a critical need for an automated, predictive system to audit water safety metrics before they reach households.

Research Question & Model Selection

The Key Question:



"Can we accurately predict the potability of a water source by classifying it as safe or unsafe for human consumption based on its physical and chemical properties?"

Research Question & Model Selection



Model Choice & Internal Benchmarking:

While Logistic Regression serves as the baseline model for binary Classification (Potable vs. Not Potable), this project will employ Internal Benchmarking against more complex algorithms to ensure maximum predictive accuracy

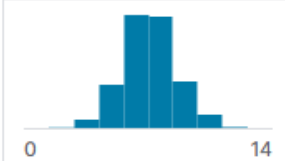
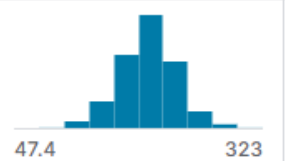
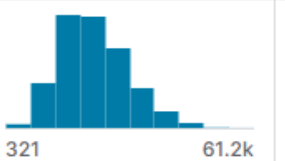
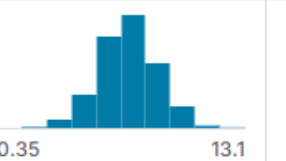
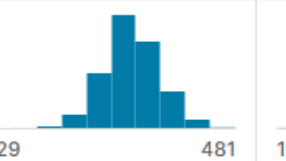
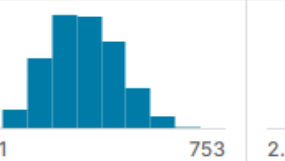
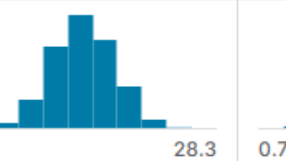
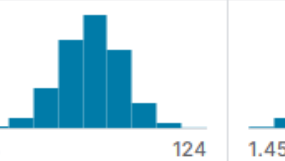
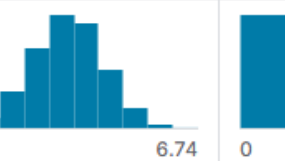
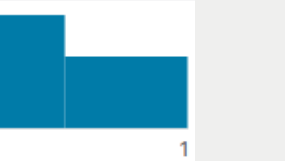
Dataset Overview



Data Source: Kaggle Water Potability Dataset.



Observations: 3,276 water samples. 

# ph decimal	# Hardness decimal	# Solids decimal	# Chloramines decimal	# Sulfate decimal	# Conductivity decimal	# Organic_carbon decimal	# Trihalomethanes decimal	# Turbidity decimal	# Potability boolean
									
014	47.4323	32161.2k	0.3513.1	129481	181753	2.228.3	0.74124	1.456.74	01
	204.8904554713363	20791.318980747026	7.300211873184757	368.51644134980336	564.3086541722439	10.3797830780847	86.9909704615088	2.9631353806316407	0
3.71608007538699	129.42292051494425	18630.057857970347	6.635245883862		592.8853591348523	15.180013116357259	56.32907628451764	4.500656274942408	0
8.099124189298397	224.23625939355776	19909.541732292393	9.275883602694089		418.6062130644815	16.868636929550973	66.42009251176368	3.0559337496641685	0
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	118.98857909025189	14285.583854224515	7.804173553073094	268.646940746221	389.3755658712614	12.70604896865791	53.928845767512236	3.5950171809576155	0
11.180284470721592	227.23146923797458	25484.50849098786	9.077200016914393	404.04163468408996	563.8854814810949	17.92780641128502	71.97660103221915	4.370561936655497	0
7.360640105838258	165.52079725952862	32452.614409143884	7.550700906704114	326.62435345560164	425.38341949538733	15.586810438033126	78.74001566430479	3.6622917828524573	0

Independent Variables (X)



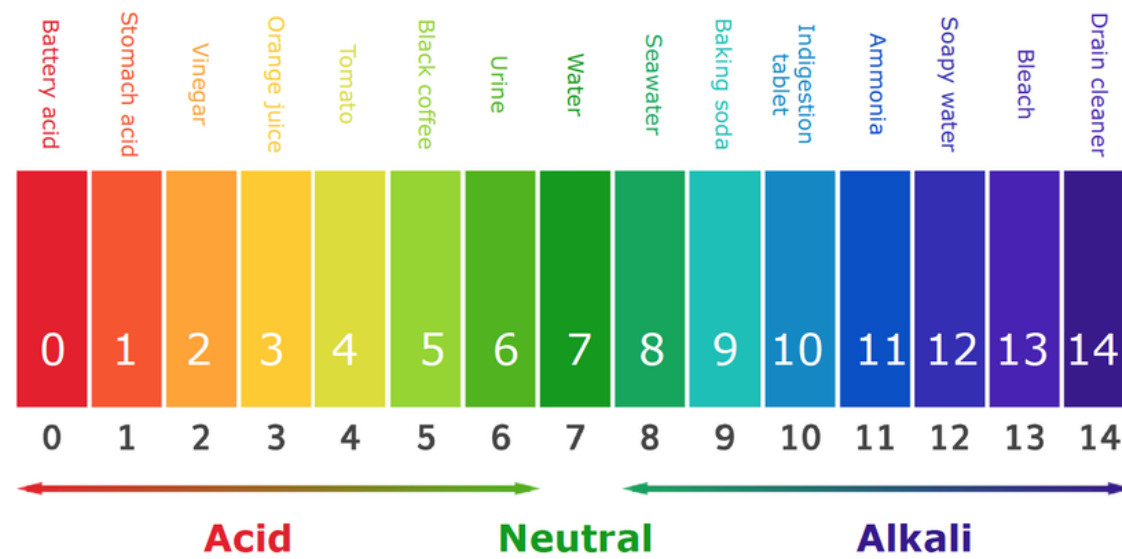
pH:
Acid-base balance
(WHO recommends 6.5–8.5).



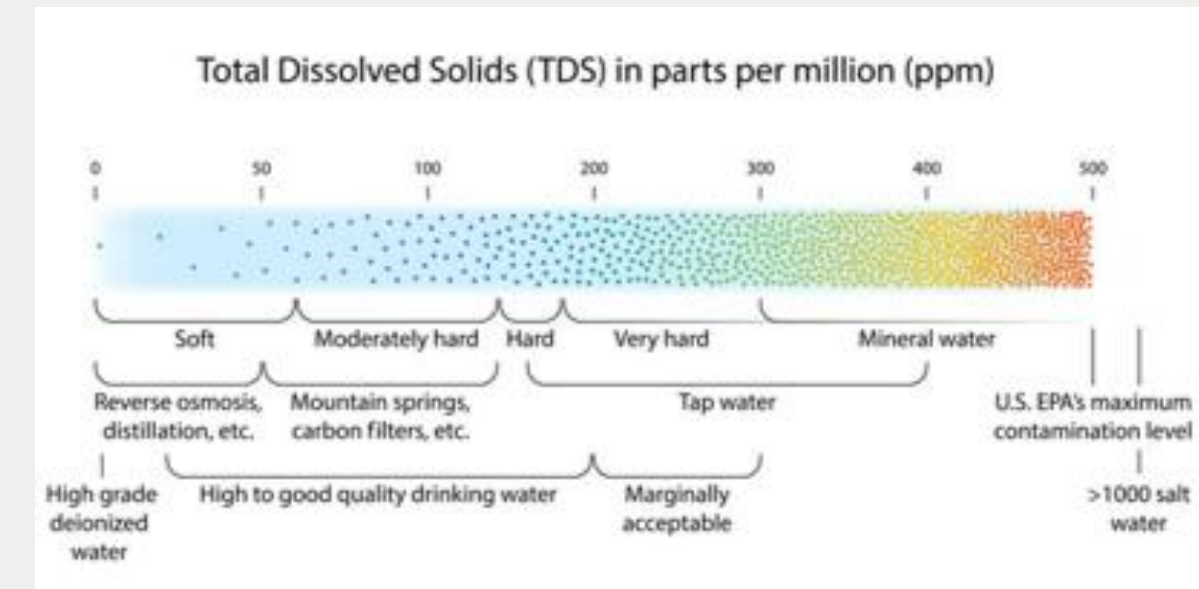
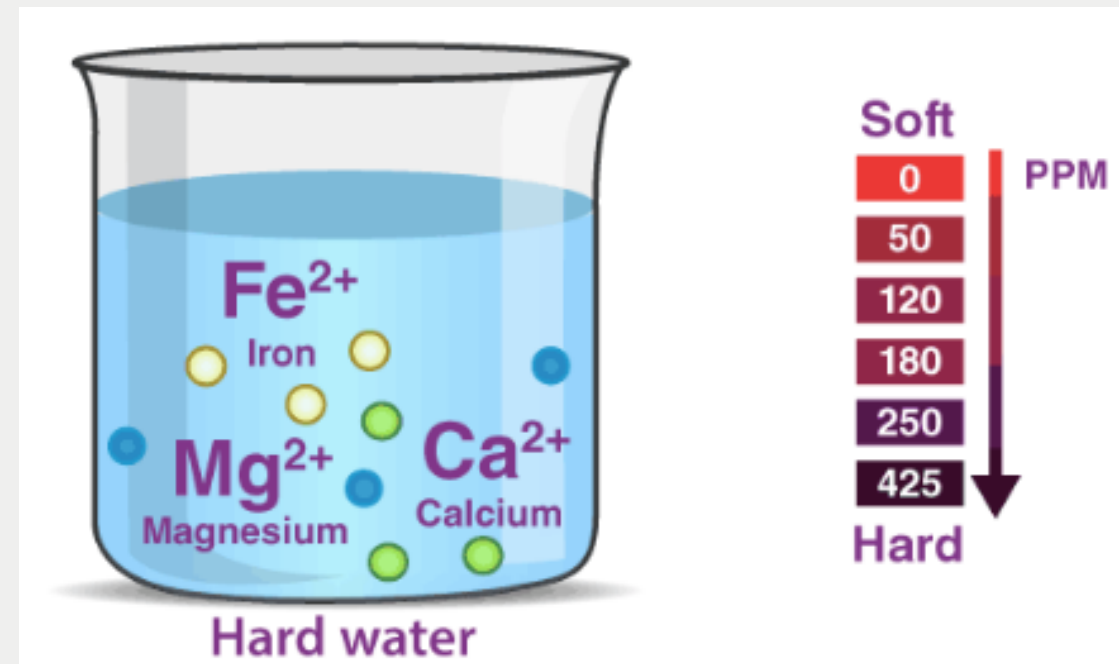
Hardness:
Mineral content
calcium and magnesium
WHO recommends: up to 300 mg/L



Solids:
Total Dissolved Solids
(TDS).
WHO recommends: up to 500 mg/L



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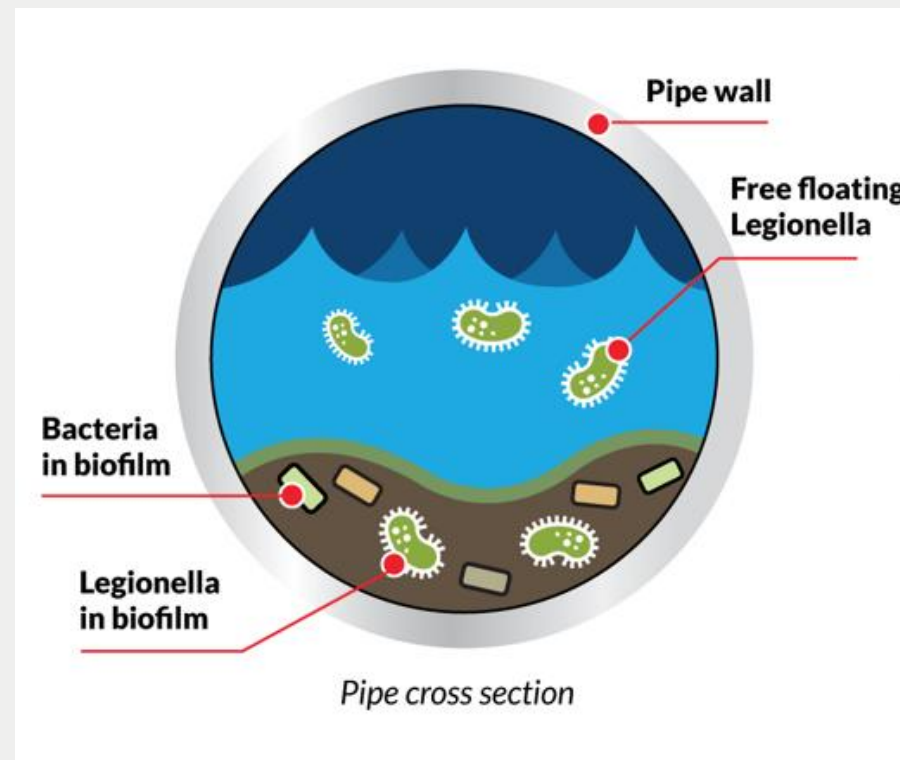
Independent Variables (X)



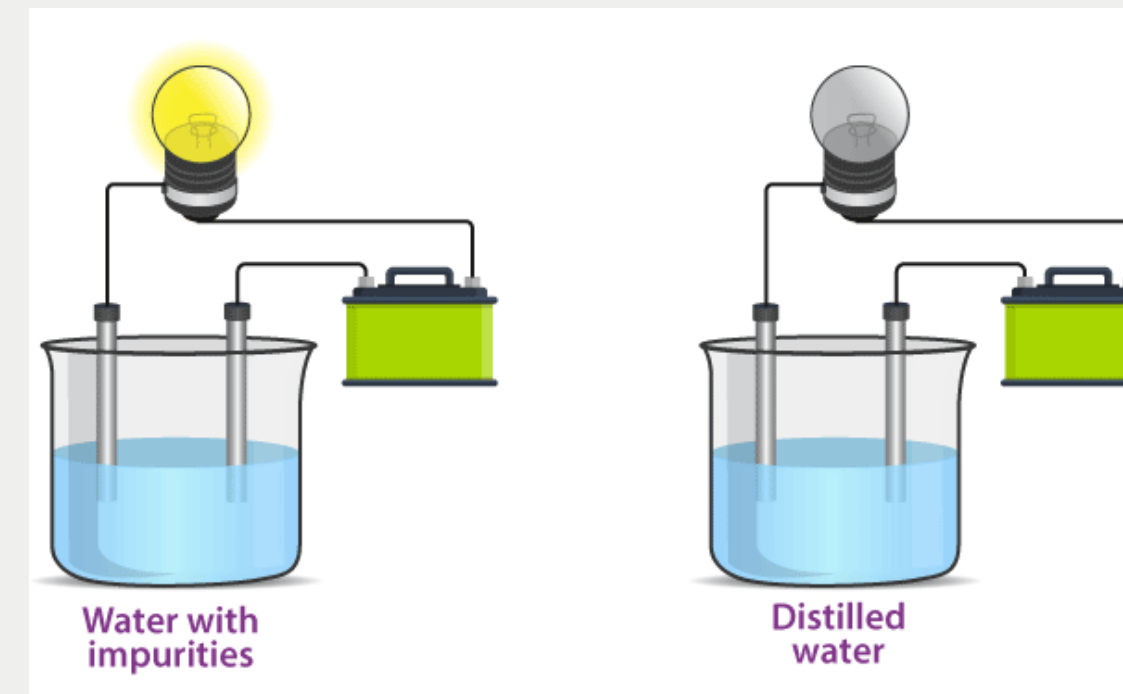
Sulfate:
Sulfate concentration
(WHO recommends: up to 250 mg/L
World Health)



Chloramines: Chloramine
concentration
(ppm)
WHO recommends: up to
4 mg/L (ppm)



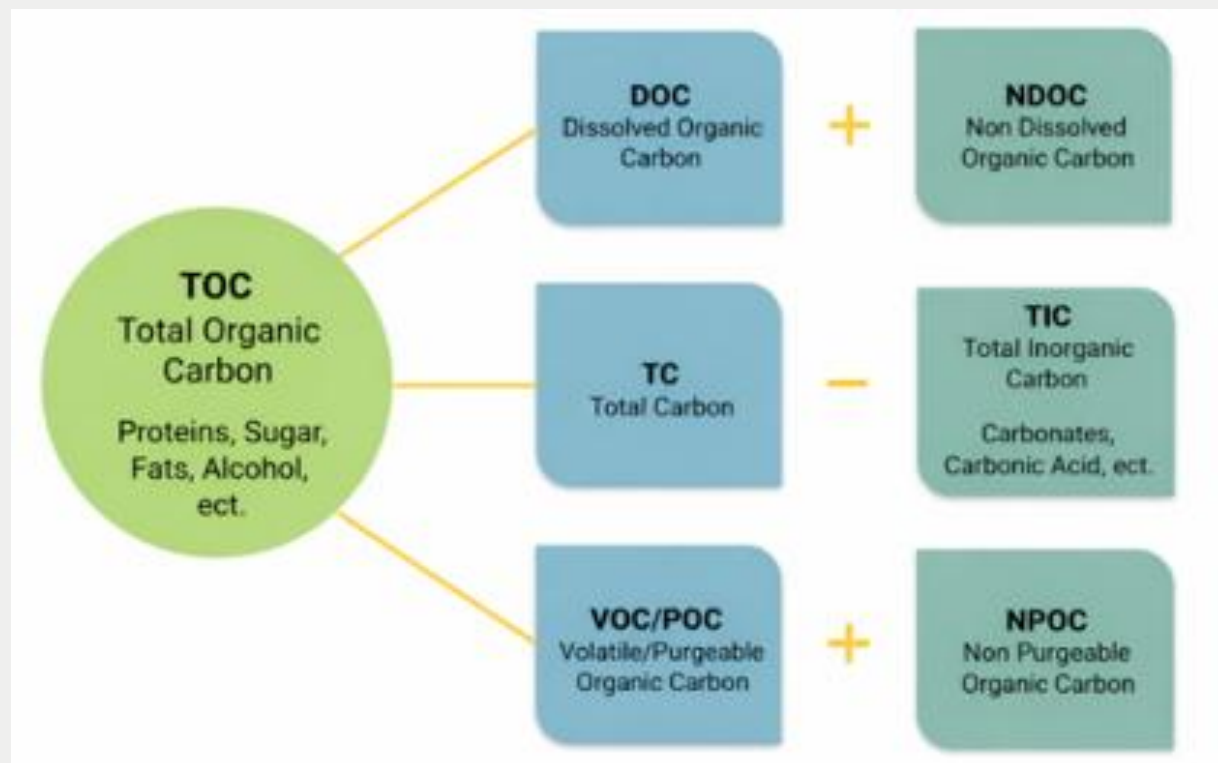
Conductivity:
Electrical conductivity
($\mu\text{S}/\text{cm}$)
WHO recommends: up to 250
mg/L World Health



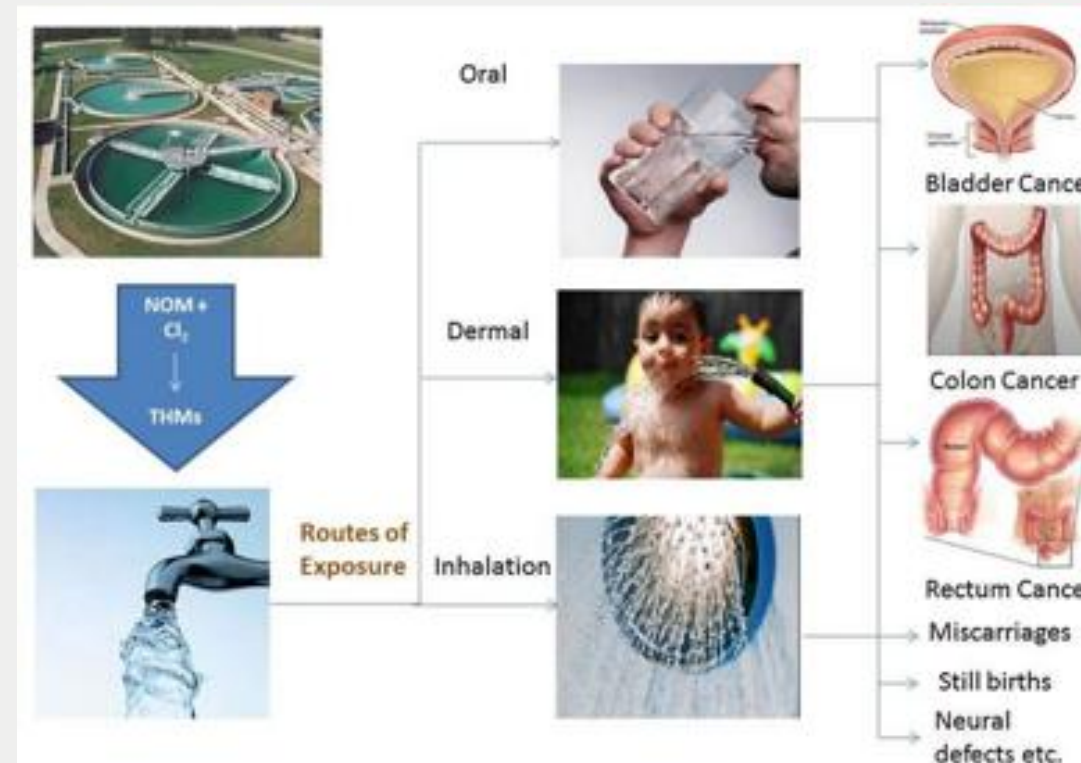
Independent Variables (X)



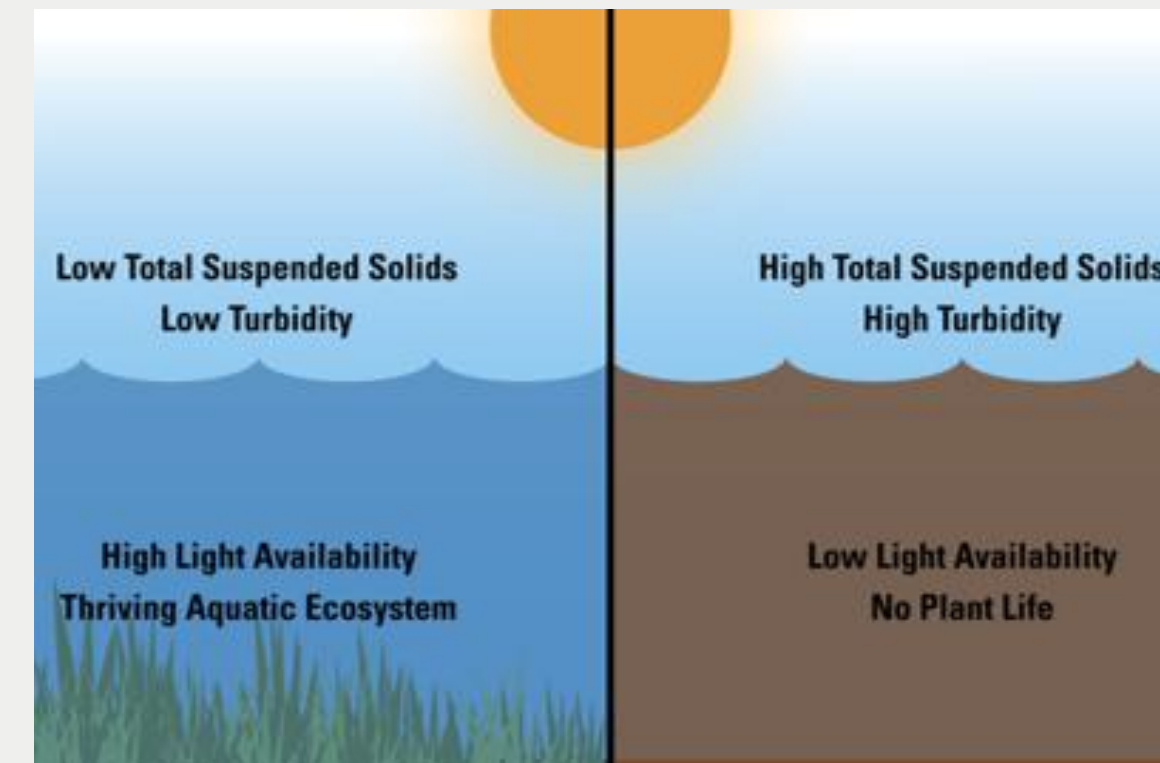
Organic Carbon:
Organic carbon content
(ppm)
Acceptable 100 ppb (0.1 ppm) to 10 ppm



Trihalomethanes:
THMs concentration
($\mu\text{g/L}$)
Should not exceed 100 $\mu\text{g/l}$



Turbidity:
Measure of light emitting
property
(NTU)
safe: < 1 NTU



Dependent Variable (Y):



Potability
(0 = Not Potable, 1 = Potable).



Potable (adjective): **suitable and safe to drink.**

Expected Social Impact



Accountability: This model serves as an independent auditing tool that holds water utility providers and Local Government Units (LGUs) accountable. By inputting reported metrics, citizens and regulators can verify if the water quality meets safety thresholds.



Decision Support: Public health officials and investors in the ESG (Environmental, Social, and Governance) sector can use this model to identify high-risk regions. Investors can use these insights to direct funding toward "green" companies that maintain superior water treatment standards, effectively penalizing entities that neglect water safety.

Statistical Evidence in the Philippine Setting

According to the Philippine Statistics Authority (PSA) and the UN-Water SDG 6 Data Portal (2024), the following statistics highlight the critical nature of water quality in the Philippines:



Safely Managed Water: As of 2024, only 48% of the population in the Philippines uses a safely managed drinking water service. This means over 60 million Filipinos still rely on water sources that may not be free from contamination or available on-premises.



Environmental Degradation: Reports indicate a 0.76% decrease in permanent surface water in the Philippines (as of 2022), increasing the concentration of pollutants in remaining water bodies.

Statistical Evidence in the Philippine Setting



Health Impact: The Department of Health (DOH) continues to monitor water-borne diseases, noting that regions with lower access to safely managed water (particularly in rural Mindanao and parts of Visayas) show higher incidences of acute gastroenteritis and typhoid.



Economic Cost: Recent literature from the OECD (2023) suggests that poor water and sanitation services cost the Philippines significant portions of its GDP due to health expenditures and lost labor productivity, emphasizing that predictive modeling is not just a technical exercise but an economic necessity.

References



Philippine Statistics Authority (2024). Highlights of the 2024 Sustainable Development Goals Pace of Progress.

https://psa.gov.ph/sdg?__cf_chl_tk=VZCOC0E4KvVGXAizYQVvrgMRqiOVIOBTYimkPfjH.qE-1776692042-1.0.1.1-15LrQ.D0hCLSZzExNEMyZZntMYgl.Xk91BvHRs6o6hk



UN–Water (2024). SDG 6 Country Profile: Philippines.

<https://www.sdg6data.org/en/country-or-area/Philippines>



WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply, Sanitation and Hygiene (2023). Progress on household drinking water, sanitation and hygiene 2000–2022.

<https://data.unicef.org/resources/jmp-report-2023/>



OECD (2023). Economic Surveys: Philippines 2023.

<https://data.unicef.org/resources/jmp-report-2023/>



Why Is The pH of Water Important?

<https://aquaclearws.com/resources/why-is-the-ph-of-water-important/>



Hardness of Water Types and Removal

<https://aquaclearws.com/resources/why-is-the-ph-of-water-important/>



Understanding Total Dissolved Solids (TDS) in Drinking Water

<https://etrlabs.com/understanding-total-dissolved-solids-tds-in-drinking-water/>



World Health Organization. (2023). Drinking Water.
OECD (2023). Economic Surveys: Philippines 2023.
<https://data.unicef.org/resources/jmp-report-2023/>



Total Organic Carbon (TOC) and its Measurement

<https://www.elgalabwater.com/blog/total-organic-carbon-toc>



Trihalomethanes in Drinking-water. Background document for development of WHO Guidelines for Drinking-water Quality

<https://www.elgalabwater.com/blog/total-organic-carbon-toc>



What Is a Turbidity Sensor? How to Choose?

<https://www.renkeer.com/what-and-how-choose-turbidity-sensor/#:~:text=The%20higher%20the%20turbidity%20in,water%20treatment%20system%20are%20faulty>



Aditya Kadiwal. (2020). Water Potability Dataset. Kaggle.
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